

Probabilistic Machine Learning

Lecture 04: Independence, Naive Bayes and Graphical Models

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The user-requested arc: naive Bayes first, as a classifier that *works*, then the independence assumption it rests on, then the general framework that replaces the assumption with a modelling choice. All units reused from the data-science course.

Module 1

Bayes as a Classifier

Guiding Question: *The data tells us how attributes behave given the class. We need the reverse. What does the conversion cost?*

Thinking With Uncertainty

Bayes' theorem lets us update a **hypothesis** H in the light of **data** D :

$$P(H \mid D) = \frac{P(H) \cdot P(D \mid H)}{P(D)}$$

This is the whole apparatus of evidence-based decision making in one line: a belief, a measurement of how well the belief explains what we saw, and a normalisation.

Speaker notes

Deliberate second pass over Bayes (P4/P5 decision rule: this is consolidation, so math first). The new element is the reading of H and D as hypothesis and data, which is the form used by every learning algorithm later in the course.



Example: Do I Have the Flu?

You have a headache and a cough. From the literature:

Likelihood

90% of people with the flu
show headache and coughing

$$P(D \mid H) = 0.9$$

Prior

5% of people currently have
the flu

$$P(H) = 0.05$$

Evidence

headache and coughing occur
in 20% of people

$$P(D) = 0.2$$

How likely is it that you have the flu?

Speaker notes

Ask for a guess first — most people say something close to 90%, repeating the base-rate error from the probability chapter. 22.5% is the answer.



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How likely is it that you have the flu?

$$P(H \mid D) = \frac{0.05 \cdot 0.9}{0.2} = 0.225$$

Why So Low?

Few people have the flu

The prior $P(H) = 0.05$ caps how far the evidence can carry us.

The symptoms are common

$P(D) = 0.2$ means many people have these symptoms *without* the flu.

A symptom is informative to the degree that it is **rare in general** but **common under the hypothesis**. A likelihood alone tells you nothing.

Two reasons, both structural. This is the criterion that reappears as feature informativeness in classification: a feature that fires for everything carries no information. Connect it explicitly — it makes information gain feel inevitable later.

Takeaway

Bayesian decision making weighs **prior × likelihood** against how ordinary the evidence is. Deciding under uncertainty means keeping all three numbers in view, not just the one the study reports.

Two Directions of the Same Connection

Observed connection

$$A_1, \dots, A_p \mid B$$

Given the class, how do the attributes behave? — the standard situation, and the one data can answer directly.

Reversed connection

$$B \mid A_1, \dots, A_p$$

Given the attributes, which class? — the diagnosis situation, and the one we actually need.

$$P(B \mid A_1, \dots, A_p) = \frac{P(B) \cdot P(A_1, \dots, A_p \mid B)}{P(A_1, \dots, A_p)}$$

This is the base-rate lesson turned into an algorithm: the data gives us the easy direction, Bayes converts it into the one we need. Ask which direction a labelled dataset supplies — attributes given class, by construction.

What the Classifier Needs

Class priors

$$P(\text{class} = y)$$

How common is each class, before seeing any attribute?

Attribute likelihoods

$$P(\text{attribute} = v \mid \text{class} = y)$$

How does each class distribute its attribute values?

Given both, the posterior $P(\text{class} \mid \text{attributes})$ follows from Bayes — and the prediction is the class with the highest posterior.

Estimating $P(A_1, \dots, A_p \mid B)$ requires the **joint** distribution of all attributes within each class: exponentially many parameters, and never enough data.

Speaker notes

End the unit on the obstacle rather than the solution — the warning is exactly the parameter explosion from the graphical-models lecture, and naive Bayes is the crudest possible independence assumption applied to it.



Module 2

Naive Bayes

Guiding Question: *Its central assumption is almost always false, and it classifies well anyway. How can both be true?*

Generative and Discriminative Classifiers

Generative

Model the **joint** distribution $P(A, B)$ — how the data was produced — then derive $P(B | A)$.

Naive Bayes.

Discriminative

Model $P(B | A)$ **directly** — only the boundary between classes.

Logistic regression.

The distinction runs through the rest of machine learning. A generative model can be sampled from and can handle missing attributes a discriminative model spends all its capacity on the decision. Neither is universally better.

The Naive Assumption

Assume the attributes are **conditionally independent given the class**:

$$P(A_1, \dots, A_p \mid B) = \prod_{j=1}^p P(A_j \mid B)$$

Exponentially many parameters collapse to p per class — one distribution per attribute.

The assumption is almost always false. Naive Bayes nonetheless classifies well, because ranking the classes correctly is easier than estimating their probabilities correctly.

Direct callback to conditional independence: this is the same move as a graphical model, taken to its extreme — a star-shaped network with the class at the centre. The note is the honest part: the probabilities are miscalibrated, the argmax often is not.

The Classifier

$$P(B_i \mid A_1, \dots, A_p) = \frac{P(B_i) \cdot \prod_{j=1}^p P(A_j \mid B_i)}{\sum_{i=1}^k P(B_i) \cdot \prod_{j=1}^p P(A_j \mid B_i)}$$

Score each class

prior × product of attribute likelihoods

Normalise

divide by the total over all classes — the theorem of total probability

Predict

the class with the largest posterior

For prediction alone the denominator can be skipped — it is the same for every class. It is needed only when the probability itself is reported.

The three-step structure is the whole algorithm. Point out that the denominator is total probability, which the students proved in the probability lecture — nothing new is being introduced here, only assembled.

Example: Enjoy Sport?

Outlook	Temperature	Humidity	Wind	EnjoySport
sunny	hot	high	weak	no
sunny	hot	high	strong	no
overcast	hot	high	weak	yes
rain	mild	high	weak	yes
rain	cool	normal	weak	yes
rain	cool	normal	strong	no

Estimate the priors and the per-attribute likelihoods by **counting**, then score a new day.

Speaker notes

Have the group compute $P(\text{yes})$ and $P(\text{sunny} \mid \text{yes})$ aloud — the estimation is pure counting, which is what makes the method so cheap. Then ask what happens for $P(\text{overcast} \mid \text{no})$: zero, which kills the entire product.



The Zero-Frequency Problem

If an attribute value never occurs with a class in the training data, its estimated likelihood is **zero** — and one zero factor drives the whole product to zero, whatever the other attributes say.

Laplace smoothing adds a pseudo-count to every combination, so that unseen does not mean impossible.

One missing count

vetoed every other piece of evidence

Naive Bayes buys tractability with a **conditional independence assumption it knows to be false** — and works anyway, because classification needs the right argmax, not the right probability.

Speaker notes

Close on the epistemic point (P3): a knowingly wrong assumption can still yield a useful model, provided you know which conclusions it does and does not support. Calibrated probabilities are exactly what it does not support.



Module 3

Independence, Properly

***Guiding Question:** Naive Bayes assumed independence to survive. What does the assumption actually claim — and when does conditioning create dependence rather than remove it?*

Conditional Independence

Definition. Events A and B are **conditionally independent given C** (written $A \perp\!\!\!\perp B \mid C$) iff

$$P(A \cap B \mid C) = P(A \mid C) \cdot P(B \mid C)$$

equivalently

$$P(A \mid B \cap C) = P(A \mid C)$$

Reading: **once C is known, B carries no further information about A .**

Math first (P5), but immediately give the reading. The second form is the one used in practice: it says B becomes redundant in the presence of C.

Dependent, but Independent Given C

Let $P(A) = \frac{8}{16}$, $P(B) = \frac{3}{16}$, $P(C) = \frac{4}{16}$ on a 16-cell sample space.

Unconditionally dependent

$$P(A \cap B) = \frac{1}{16}$$

$$P(A) \cdot P(B) = \frac{3}{32}$$

$$\frac{1}{16} \neq \frac{3}{32}, \text{ so } A \not\perp B$$

Conditionally independent given C

$$P(A \cap B \mid C) = \frac{1}{4}$$

$$P(A \mid C) \cdot P(B \mid C) = \frac{2}{4} \cdot \frac{2}{4} = \frac{1}{4}$$

$$\text{so } A \perp B \mid C$$

Also directly: $P(A \mid B \cap C) = \frac{1}{2} = P(A \mid C)$.

The whole idea in one comparison. Conditioning on C *created* independence that did not exist before. Ask for a real example: ice-cream sales and drownings are dependent, but conditionally independent given the temperature.

Why This Matters

Explaining away

A common cause C can make two effects look related.

Model size

Conditional independence lets a joint distribution factorise into small pieces.

Later in the course

Naive Bayes assumes it.
Graphical models are built from it.

Conditional independence neither implies nor is implied by ordinary independence. Both directions can fail.

Flag forward links without using them: students only need to know that this definition is the hinge for models later. The warning box is worth a moment — students routinely assume one form gives the other.

Takeaway

Conditioning can **create** independence as easily as it destroys it. "Given what I already know, does this extra fact still tell me anything?" is the question that makes large probabilistic models tractable.

Module 4

From Assumption to Model

Guiding Question: *Naive Bayes assumes everything is independent given the class. What if we could state exactly which independences hold?*

The Problem: Joint Distributions Explode

Consider a joint distribution $p(x_1, \dots, x_n)$ over n **binary** random variables. How many configurations are there?

Guiding question of the module. 30 binary variables already exceed a billion parameters. Ask what would have to be true of the world for the table to be compressible — the answer is independence, which the students already have.

The Problem: Joint Distributions Explode

Consider a joint distribution $p(x_1, \dots, x_n)$ over n **binary** random variables. How many configurations are there?

2^n states, $2^n - 1$ free parameters

Specification

For large n , writing down all values is impossible.

Inference

A single marginal $p(x_1)$ requires summing over 2^{n-1} states.

Chain rule?

It rewrites the joint exactly — but does not reduce the number of values.

What a Graphical Model Is

Definition (Graphical model). A graphical model depicts the dependence and independence relations of a distribution using a graph G as its underlying structure.

Modelling

Describe the variables of a domain and their interactions — how $p(x_1, \dots, x_n)$ **factorises**.

Inference

Answer queries such as $p(x_1 \mid x_2 = a, \dots)$ efficiently over that factorisation.

Usually one class of model is used for modelling and another for inference the latter is often compiled from the former.

Two purposes, two different graph classes — worth stating now so that Markov networks later do not look like a competing formalism but a compilation target.

How a Graph Can Help

Probabilistic inference is summing and multiplying. Let every variable be a node, and read edges as interaction:

No edge

$$f(a, b) = g(a) \cdot h(b)$$

The variables are independent — the factor splits.

Edge

$$f(a, b) \neq g(a) \cdot h(b)$$

The variables are dependent — the factor does not split.

Edge **direction** can encode further structure. The principle extends to any number of variables.

Graphical models have limited expressive power: not every conditional independence statement can be drawn, and different graphs can represent the same distribution.

The honest caveat matters — students should not believe the graph captures everything. "The graph is a picture of a factorisation" is the accurate slogan.

Example: Why Is the Grass Wet?

Tracey leaves her house and sees that her grass is wet. **Was it rain, or did she leave the sprinkler on?** She notices her neighbour Jack's grass is also wet.

R

raining (1) or not (0)

S

Tracey's sprinkler on
(1) or off (0)

J

Jack's grass wet (1)
or dry (0)

T

Tracey's grass wet
(1) or dry (0)

Barber's classic example. Ask students to predict, before any arithmetic, whether Jack's wet grass makes the sprinkler more or less likely — this sets up explaining away, and intuitions differ, which is the point.

Building the Model

The full joint over four binary variables needs $2^4 - 1 = 15$ probabilities. The chain rule rewrites it but does not shrink it:

$$p(T, J, R, S) = p(T \mid J, R, S)p(J \mid R, S)p(R \mid S)p(S)$$

Now bring in **background knowledge** — assumptions of conditional independence:

Tracey's grass does not depend on Jack's

$$p(T \mid J, R, S) = p(T \mid R, S)$$

Jack's grass does not depend on Tracey's sprinkler

$$p(J \mid R, S) = p(J \mid R)$$

Rain is not influenced by the sprinkler

$$p(R \mid S) = p(R)$$

$$p(T, J, R, S) = p(T \mid R, S)p(J \mid R)p(R)p(S)$$

Only $4 + 2 + 1 + 1 = 8$ probabilities remain — and each one is a question a person can actually answer.

The heart of the module: the reduction is not mathematical trickery, it is domain knowledge written as conditional independence. The second half of the callout matters as much as the first — the factors are locally interpretable.

The Probability Tables

Factor	$p(\cdot)$	$1 - p(\cdot)$
$p(R = 1)$	0.2	0.8
$p(S = 1)$	0.1	0.9
$p(J = 1 \mid R = 0)$	0.2	0.8
$p(J = 1 \mid R = 1)$	1.0	0.0
$p(T = 1 \mid R = 1, S = 0)$	1.0	0.0
$p(T = 1 \mid R = 1, S = 1)$	1.0	0.0
$p(T = 1 \mid R = 0, S = 1)$	0.9	0.1
$p(T = 1 \mid R = 0, S = 0)$	0.0	1.0

Missing edges in the graph are exactly the assumptions above the tables supply the numbers.

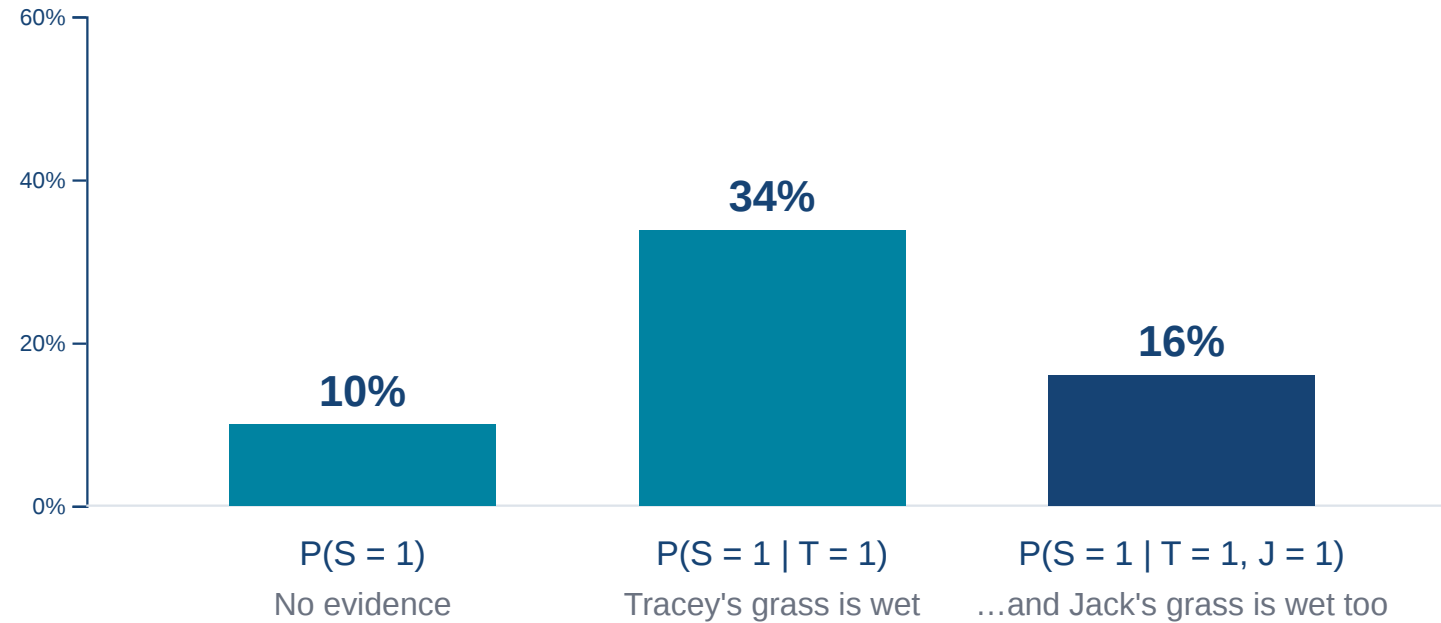
Speaker notes

Point out that the last row encodes "grass does not get wet by itself" and the 1.0 entries encode "rain always wets grass" — modelling choices, visible and challengeable, which is the epistemic virtue of the approach (P3).



Inference: What Does the Evidence Say?

Rain explains the wetness — the sprinkler falls back towards its prior



Speaker notes

Stepped viz (P6). Step 1: wet grass raises $P(\text{sprinkler})$ from 10% to 33.8%. Step 2: Jack's grass is wet too — rain now explains the wetness, and the sprinkler falls to 16.0%: most of the way back, but still above the 10% prior, because the sprinkler remains a possible additional cause. Ask for the prediction before each click. Computed by exact enumeration of the factorised joint, i.e. marginalisation the students already know.



Explaining Away

$$p(S = 1 \mid T = 1) = \frac{\sum_{J,R} p(T=1 \mid R, S=1)p(J \mid R)p(R)p(S=1)}{\sum_{J,R,S} p(T=1 \mid R, S)p(J \mid R)p(R)p(S)}$$

Two independent causes (R and S) become **dependent once their common effect is observed**. Learning that rain occurred lowers the probability of the sprinkler, even though rain and sprinkler are unconditionally independent.

This is the conditional-independence unit returning with a vengeance: conditioning on a common effect *creates* dependence — the opposite of the common-cause case seen earlier. Name it "explaining away" it is the single most useful piece of vocabulary in this module.

Takeaway

A graphical model turns **background knowledge into a factorisation**: missing edges are conditional independence assumptions, and each one removes parameters. The graph is a picture of how the joint splits — nothing more, and nothing less.